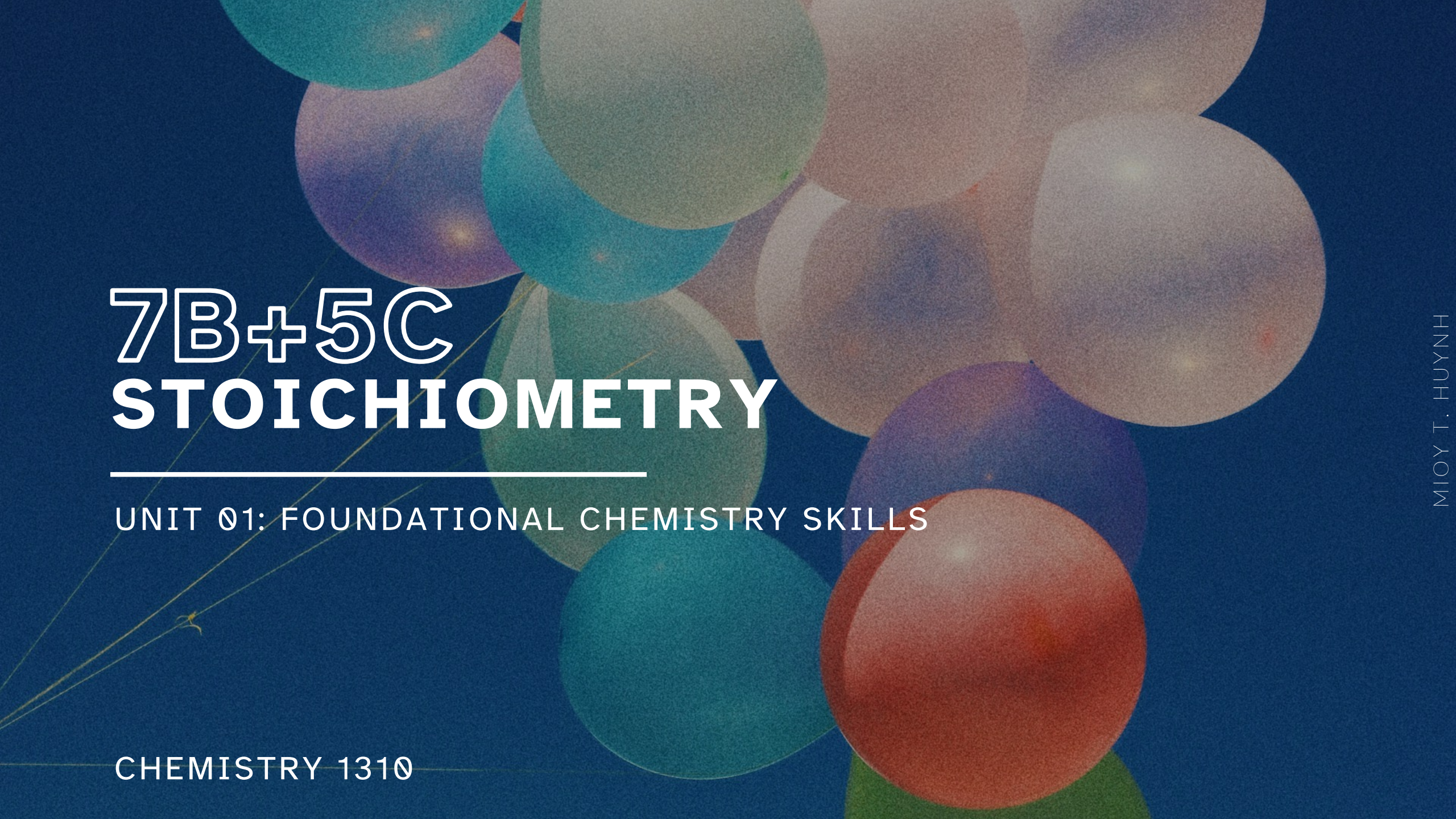




7B+5C STOICHIOMETRY

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 7

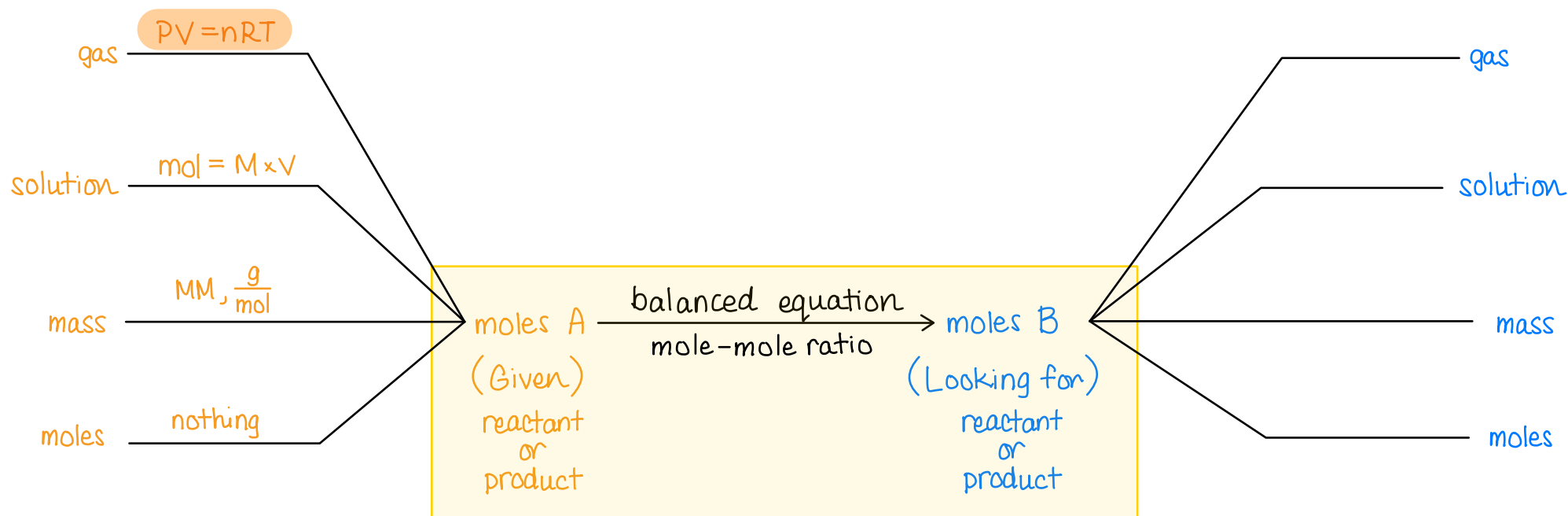
1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using the quantities of the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Predict how the pressure in a fixed-volume container will change after a complete reaction.
5. Use the kinetic molecular theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.

Gases in Chemical Reactions

7.9

Gaseous stoichiometry

- Recall, stoichiometry is *always* about moles.
- For gases, use the ideal gas law to determine the moles of gases present.



ICQ 7B+5C

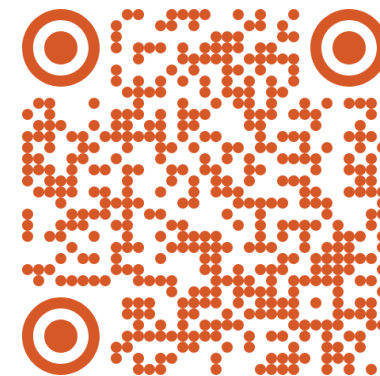
Stoichiometry Review

You can earn full credit for **ICQ 7B+5C** in one of two ways:

1

Attendance:

- a) Tap Buzzcard at the front of room.
- b) Work on the problems in class.
 - Physical copies available.
 - Download a copy by scanning QR code or go to:
gatech.box.com/v/CHEM1310stoichiometry
- c) I will update your score manually to 4/4 on Friday, January 31.



2

Attempt **ICQ 7B+5C** via Canvas Quizzes.

IN-CLASS QUESTION 1

TIP This is a LR problem because quantities of ALL reactants provided (i.e. they're limited)

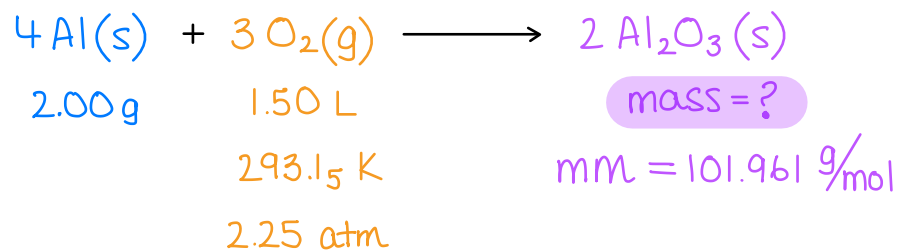
Learning Objective(s): 7.4

$$mm = 26.982 \text{ g/mol}$$

A 2.00 g sample of aluminum metal is combined with 1.50 L of oxygen gas at 20.0 °C and 2.25 atm. What mass (in grams) of solid aluminum oxide can be formed?

293.15 K

Write chemical equation.



Al³⁺, O²⁻

Identify limiting reactant (LR).

$$2.00 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.982 \text{ g Al}} \times \frac{2 \text{ mol Al}_2\text{O}_3}{4 \text{ mol Al}} = 0.03706 \text{ mol Al}_2\text{O}_3 \longrightarrow \text{Al is LR!}$$

$$n_{\text{O}_2} = \frac{P_{\text{O}_2} V}{RT} = \frac{(2.25 \text{ atm})(1.50 \text{ L})}{\left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(293.15 \text{ K})} = 0.1403 \text{ mol O}_2 \times \frac{2 \text{ mol Al}_2\text{O}_3}{3 \text{ mol O}_2} = 0.09353 \text{ mol Al}_2\text{O}_3$$

Determine mass Al₂O₃ from LR and stoichiometry.

$$0.03706 \text{ mol Al}_2\text{O}_3 \times \frac{101.961 \text{ g Al}_2\text{O}_3}{1 \text{ mol Al}_2\text{O}_3} = 3.78 \text{ g Al}_2\text{O}_3$$

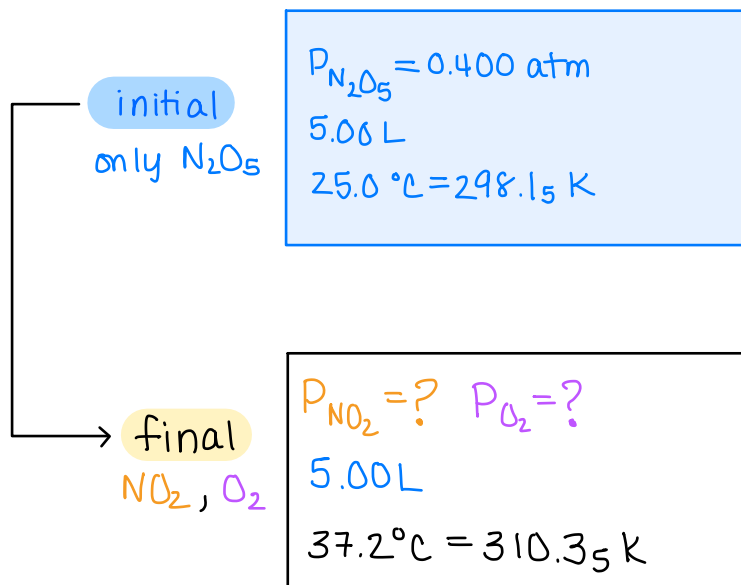
IN-CLASS QUESTION 2

Learning Objective(s): 7.4

N_2O_5 is an unstable gas that decomposes: $2 \text{N}_2\text{O}_5(\text{g}) \rightarrow 4 \text{NO}_2(\text{g}) + \text{O}_2(\text{g})$

A sample of N_2O_5 is placed into a 5.00 L container at 25.0 °C and an initial pressure of 0.400 atm. After the entire N_2O_5 sample decompose, the temperature is 37.2 °C. What is the partial pressure of NO_2 (P_{NO_2} in atm) in the container after decomposition?

Stoichiometry: find $n_{\text{N}_2\text{O}_5}$, then n_{NO_2} , then P_{NO_2} using ideal gas law



$$n_{\text{N}_2\text{O}_5} = \frac{P_{\text{N}_2\text{O}_5} V}{RT} = \frac{(0.400 \text{ atm})(5.00 \text{ L})}{\left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right)(298.15 \text{ K})} = 0.08174 \text{ mol N}_2\text{O}_5 \text{ initially}$$

$$n_{\text{NO}_2} = 0.08174 \text{ mol N}_2\text{O}_5 \times \frac{4 \text{ mol NO}_2}{2 \text{ mol N}_2\text{O}_5} = 0.1635 \text{ mol NO}_2 \text{ made}$$

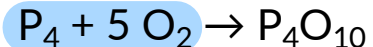
$$P_{\text{NO}_2} = \frac{n_{\text{NO}_2} RT}{V} = \frac{(0.1635 \text{ mol}) \left(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}\right) (310.35 \text{ K})}{5.00 \text{ L}} = 0.833 \text{ atm}$$

EXAMPLE PROBLEM

Learning Objective(s): 5.3

Clue to identify LR problem: Quantities of ALL reactants provided.
i.e. They're limited/finite, so one might run out.

Each question statement below is a stoichiometry problem. For each, determine if you would first need to identify a limiting reactant (LR) or not.

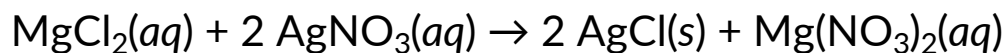


Burning phosphorus in oxygen produces tetraphosphorus decoxide. What is the percent yield of the reaction if 10.0 g of P_4O_{10} are obtained at the end of the reaction between 0.200 mol P_4 and 0.200 mol O_2 ?

☒ Yes, I would need to identify a LR.

☐ No, I do not need to identify a LR.

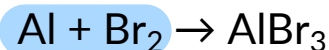
What mass (in grams) of MgCl_2 is needed to react completely with 10.0 g of AgNO_3 ?



☐ Yes, I would need to identify a LR.

☒ No, I do not need to identify a LR.

Consider the reaction between 10.0 g Al and 10.0 g Br_2 according to the unbalanced equation. If 6.97 g of AlBr_3 is obtained in the lab, what is the percent yield?



☒ Yes, I would need to identify a LR.

☐ No, I do not need to identify a LR.

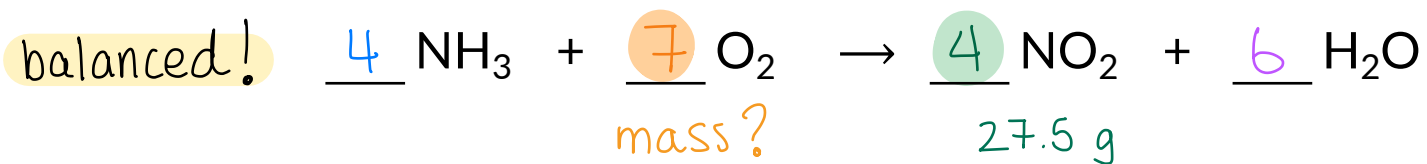
EXAMPLE PROBLEM

Learning Objective(s): 5.2

$$mm = 31.998 \text{ g/mol}$$

$$mm = 46.005 \text{ g/mol}$$

What mass (in grams) of oxygen is required to produce 27.5 g of NO_2 according to the unbalanced chemical equation below?



mass NO_2 $\xrightarrow{\text{mm}}$ moles NO_2 $\xrightarrow{\text{mole:mole ratio}}$ moles O_2 $\xrightarrow{\text{mm}}$ mass O_2

$$27.5 \text{ g NO}_2 \times \frac{1 \text{ mol NO}_2}{46.01 \text{ g NO}_2} = 0.5977 \text{ mol NO}_2 \times \frac{7 \text{ mol O}_2}{4 \text{ mol NO}_2} = 1.046 \text{ mol O}_2 \times \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} = 33.5 \text{ g O}_2$$

TIP This is a LR problem because quantities of ALL reactants provided (i.e. they're limited)

EXAMPLE PROBLEM

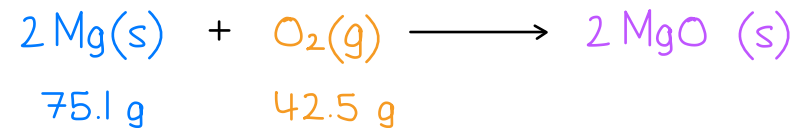
Learning Objective(s): 5.3

$$\text{mm} = 31.998 \text{ g/mol}$$

$$\text{mm} = 24.305 \text{ g/mol}$$

If 42.5 g of O_2 reacts with 75.1 g of Mg to produce MgO, then what mass (in grams) of the excess reagent remains?

Write chemical equation.



Identify limiting reactant (LR).

$$75.1 \text{ g Mg} \times \frac{1 \text{ mol Mg}}{24.305 \text{ g Mg}} \times \frac{2 \text{ mol MgO}}{2 \text{ mol Mg}} = 3.086 \text{ mol MgO}$$

$$42.5 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{31.998 \text{ g O}_2} \times \frac{2 \text{ mol MgO}}{1 \text{ mol O}_2} = 2.629 \text{ mol MgO} \longrightarrow \text{O}_2 \text{ is LR; Mg is xs!}$$

Determine mass Mg used from LR and stoichiometry.

$$42.5 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{31.98 \text{ g O}_2} \times \frac{2 \text{ mol Mg}}{1 \text{ mol O}_2} \times \frac{24.305 \text{ g Mg}}{1 \text{ mol Mg}} = 64.56 \text{ g Mg used}$$

Determine mass Mg left.

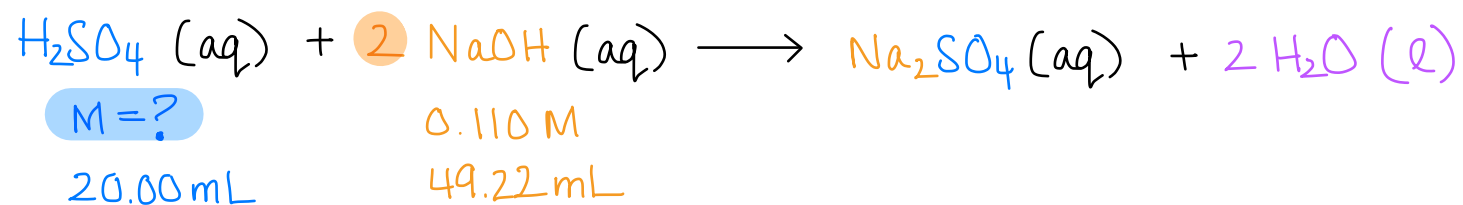
$$\begin{array}{ccc} 75.1 \text{ g Mg} & - & 64.56 \text{ g Mg} = 10.5 \text{ g Mg left/remaining} \\ \text{initial} & & \text{used} \end{array}$$

EXAMPLE PROBLEM

Learning Objective(s): 5.5

A 20.00 mL solution of H_2SO_4 of unknown concentration reacts with 0.100 M NaOH.

You find that 49.22 mL of 0.110 M NaOH is required for *complete* neutralization (100 % yield). What is the initial concentration (in molarity, M) of the original H_2SO_4 solution?



$$0.04922 \text{ L NaOH} \times \frac{0.110 \text{ mol NaOH}}{1 \text{ L}} = 0.005414 \text{ mol NaOH} \times \frac{1 \text{ mol H}_2\text{SO}_4}{2 \text{ mol NaOH}} = 0.002707 \text{ mol H}_2\text{SO}_4$$

$$[\text{H}_2\text{SO}_4] = \frac{0.002707 \text{ mol H}_2\text{SO}_4}{0.02000 \text{ L}} = 0.135 \text{ M H}_2\text{SO}_4$$